

ICT CSA Tutorial 03

- 01) Explain the difference between Application Software and System Software, providing one example for each.
- 02) Convert the decimal number **37.0625** to its binary equivalent. Show your work for both the integer and fractional parts.
- 03) What are the two main parts of an instruction, and what is the function of each part?
- 04) List the five functional groups into which MIPS32 instructions can be classified.
- 05) According to the slides, what is the requirement for word alignment in MIPS, and what is the benefit of this requirement?
- 06) MIPS32 is a load-store architecture. What does this mean for how operations are performed and how main memory is accessed?
- 07) Explain the concept of **Direct Addressing** and provide one of its key disadvantages.
- 08) What is the main characteristic of Immediate Addressing, and why is it considered fast but limited in range?
- 09) When two 32-bit numbers are multiplied in MIPS, what is the size of the product and where are the two halves of the result stored?
- 10) What is the purpose of an Instruction Set Architecture (ISA), and what are some of the key components it typically consists of?